

SHAMIR KHAN

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Education

Bachelor of Science in Computer Science

Wilfrid Laurier University

Expected April 2028

Waterloo, ON

Relevant Coursework: Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Engineering, Database Systems, Computer Networks, Systems Architecture, Computer Architecture, Discrete Mathematics

Technical Experience

Operations and Technical Lead

August 2026

The J Healthcare Initiative (via Riipen Labs)

Remote, ON

- **Led technical discovery and digital audits** for a healthcare non-profit, analyzing user journeys and information architecture (IA) to eliminate critical navigation bottlenecks.
- **Architected a self-guided triage workflow** and restructured intake funnels, cutting resource discovery pathways to **under 3 clicks** to reduce user bounce rates and cognitive friction.
- **Authored a 30–60–90 day technical execution roadmap** with **\$0 additional tech spend**, aligning native web tooling and non-profit grant allocations across an 11-slide client delivery brief.

Full-Stack Developer and Team Lead

April 2026

ConHacks – InstantService

Waterloo, ON

- **Led a team of 4 developers** to architect an AI-native mobile-first Progressive Web Application (PWA) using **Gemini 1.5 Pro**, reducing client service-acquisition time by **70%** via automated natural language request classification.
- **Engineered a multi-weighted contractor matching algorithm** with **FastAPI** and **Snowflake**, optimizing contractor allocation to drive a **40% increase in job volume** and deliver **instant, under-3-minute service confirmations**.
- **Architected secure backend infrastructure** for tier-based pricing and contractor verification, automating verification of license/insurance certificates to establish a high-trust production environment.

High-Impact Projects

CloudGuard – Cloud Security Posture Management (CSPM) | [Live Site](#) | [GitHub](#)

July 2026

- **Architected an event-driven distributed microservices platform** (C++, Python, Java, Scala) containerized with **Docker** and deployed on **Render** and **Vercel**, achieving **99.9% uptime** with a persistent **PostgreSQL** backend.
- **Built a native C++ real-time telemetry agent** using **Windows APIs** (GlobalMemoryStatusEx, GetSystemTimes) and Linux **/proc** parsers, streaming hardware metrics to **Apache Kafka** topics at **5-second intervals**.
- **Developed a Python IsolationForest ML anomaly pipeline** processing live Kafka streams to detect SSH brute-force intrusions and abnormal cost surges, cutting security MTTR by **85%**.
- **Engineered a Scala Spark streaming analytics engine** aggregating real-time telemetry across **AWS, Azure, and GCP** into sliding time windows to correlate distributed billing anomalies with sub-second latency.

Subscription Manager – Scalable Full-Stack REST API | [GitHub](#)

March 2026

- **Built a production-grade scalable RESTful API** using **Node.js, Express, and MongoDB**, enforcing **JWT authentication**, strict **Zod** schemas, and automated unit testing to eliminate **95%** of unvalidated payload errors.
- **Hardened API defense-in-depth** by integrating **Arcjet** bot protection and IP rate limiting, sustaining **99.9% uptime** against automated scraping and DDoS vectors on **Render**.
- **Automated scheduled lifecycle notifications** and trial expiration workflows with high fault-tolerance using **Upstash Workflow** serverless queues and **Nodemailer**.

MSA Platform – Digital Community Web Application | [Live Site](#)

Jan 2026

- **Improved navigation flows and accessibility** for a **250+ member community** platform, architecting responsive UI/UX layouts for the official **React/Next.js** digital ecosystem.
- **Developed interactive event registration modules** using **TypeScript** and **Payload CMS**, supporting community-driven initiatives that generated **\$50K+** in charitable donations.

Technical Skills Summary

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL, HTML/CSS, Scala, Kotlin

Frameworks & Libraries: React.js, Next.js, Node.js, Express, FastAPI, Spring Boot, Apache Spark, Tailwind CSS

Developer Tools & Cloud: Docker, Apache Kafka, Snowflake, PostgreSQL, MongoDB, Git, GitHub Actions, CI/CD Pipelines, Linux/Unix, Render, Vercel

Core Concepts: Computer Science Fundamentals, Software Development Life Cycle (SDLC), Agile Methodologies, Object-Oriented Programming (OOP), Systems Architecture, Unit Testing, Distributed Systems, Real-Time Technology, Mobile Web Development (PWA), RESTful APIs, JWT Authentication